

SSTO fusion powered spaceplane using CHARM architecture with $p\text{-}^{11}\text{B}$ fuel

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Abstract

We specify an SSTO spaceplane that flies Space Shuttle-style operations—including a Shuttle-class cargo bay—from a municipal airport to ISS altitude LEO, powered by a continuous CHARM $p\text{-}^{11}\text{B}$ plant with direct energy conversion (DEC). Flight regimes use free air, then scarce air, then carried water. Each design step is written as a closed set of equations. We guesstimate the reactor mass hole, constrain water as a function of dry mass and vacuum Δv , impose a 1 GW plant with space restart and DEC, and solve a reference all-up mass. Combined-cycle engine maps and CHARM size/performance constraints follow.

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1 Design goals (the plane)

ID	Goal	Statement
G1	SSTO	Single stage from runway to ISS-class LEO; no discarded boosters or external tank
G2	Shuttle style	Orbiter-like airframe: wing-body, TPS, runway landing, gear, control surfaces; crew systems (toilet, ECLSS, food)
G2b	Single-deck crew	Flatten Shuttle cabin to one deck: flight deck; six reclining seats; O ₂ /N ₂ + ECLSS; luggage; forward/side ground door; large aft airlock into bay
G2c	Length	Stretch OML as needed for airlock, ECLSS tanks, battery, and fusion-fuel tanks—reference $L \approx 52$ m
G3	Shuttle cargo bay	Usable bay ≈ 18.3 m $\times$ 4.6 m class for payload—not filled with reactors
G4	Payload	Shuttle-class cargo: $m_{\text{pl}} = 24\,400$ kg reference
G5	Destination	Circular LEO compatible with ISS altitude (≈ 400 km); plane-change to 51.6° treated as margin
G6	Municipal airport	Takeoff/landing on long civil runways; first gear cool enough for noise/oversight; no vertical pad
G7	One engine	Single combined-cycle propulsion string; deadstick glide if plant fails
G8	Clean fuel	$p + {}^{11}\text{B} \rightarrow 3\alpha + 8.7$ MeV; CHARM bottle; DEC-first electricity
G9	Power	Plant electrical bus peak $P_\star = 1$ GW (design target)

Not goals: expendable stages; D–T breeding plant; filling the bay with the fusion island.

2 Constants and symbols

Symbol	Meaning	Reference value
R_E	Earth radius	6.371×10^6 m

Symbol	Meaning	Reference value
μ_E	GM_E	$3.986 \times 10^{14} \text{ m}^3/\text{s}^2$
h_{ISS}	ISS-class altitude	$4.00 \times 10^5 \text{ m}$
r	Orbit radius	$R_E + h_{\text{ISS}}$
g_0	Standard gravity	9.80665 m/s^2
m_{pl}	Cargo-bay payload	$2.44 \times 10^4 \text{ kg}$
$P_\star$	CHARM bus peak power	$1.00 \times 10^9 \text{ W}$

Masses (all kg):

Symbol	Meaning
m_{af}	Airframe primary: fuselage, wings, TPS (excl. gear/controls)
m_{gear}	Landing gear (nose + mains), doors, actuators
m_{ctrl}	Control surfaces + actuators (elevons, rudder, speed brake / body flap equiv.)
m_{crew}	Crew cabin systems: ECLSS, O ₂ /N ₂ tanks, pressure control, toilet, galley, food, luggage, airlock fittings
m_{str}	$m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}}$
m_{eng}	Combined-cycle engine + inlets + nozzles + ducts
m_{C}	CHARM island (chambers, magnets, RF, DEC, local shield, cryo, bath)
m_{bat}	Flight battery / APU (restart + hotel)
m_{f}	$p\text{-}^{11}\text{B}$ fuel
m_{w}	Water carried at takeoff (vacuum reaction mass)
m_{dry}	All mass except water
m_0	Gross liftoff mass (GLOW)
m_{ins}	Mass at LEO insertion (after water burn)

3 MWh budget: Shuttle-class mass to ISS LEO

3.1 Ideal orbital specific energy

$$r = R_E + h_{\text{ISS}}, \quad v_{\text{orb}} = \sqrt{\frac{\mu_E}{r}}, \quad \varepsilon_{\text{orb}} = -\frac{\mu_E}{2r}, \quad \varepsilon_{\text{surf}} \approx -\frac{\mu_E}{R_E}.$$

$$\Delta\varepsilon = \varepsilon_{\text{orb}} - \varepsilon_{\text{surf}} = \frac{\mu_E}{R_E} - \frac{\mu_E}{2r} \approx 3.31 \times 10^7 \text{ J/kg} = 33.1 \text{ MJ/kg}.$$

Numerical anchors:

$$v_{\text{orb}} \approx 7.67 \text{ km/s}, \quad \Delta\varepsilon \approx 9.19 \text{ kWh/kg}.$$

3.2 Orbital energy of the inserted vehicle

Everything that arrives at ISS altitude is still aboard (SSTO):

$$E_{\text{orb}} = m_{\text{ins}} \Delta \varepsilon.$$

In MWh:

$$E_{\text{orb}}^{(\text{MWh})} = m_{\text{ins}} \cdot \frac{\Delta \varepsilon}{3.6 \times 10^9}.$$

m_{ins}	E_{orb}	E_{orb}
100 t (orbiter+cargo class)	3.31 TJ	920 MWh
150 t	4.97 TJ	1380 MWh
200 t	6.62 TJ	1840 MWh

3.3 Source energy (plant must supply)

Air-breathing and rocket/plasma paths leave energy in the wake and fight drag/gravity. Define a mission multiplier $\kappa_E \geq 1$:

$$E_{\text{src}} = \kappa_E E_{\text{orb}} = \kappa_E m_{\text{ins}} \Delta \varepsilon.$$

Working band for planning:

$$\kappa_E \in [2, 4] \Rightarrow E_{\text{src}}^{(\text{MWh})} \approx (18\text{--}37) m_{\text{ins},100\text{t}}$$

i.e. about 1.8–3.7 GWh of source energy per 100 t inserted, scaling linearly with m_{ins} .

Time at constant bus power $P_\star$:

$$t_{\text{src}} = \frac{E_{\text{src}}}{P_\star} = \frac{\kappa_E m_{\text{ins}} \Delta \varepsilon}{P_\star}.$$

For $m_{\text{ins}} = 1.9 \times 10^5$ kg, $\kappa_E = 3$, $P_\star = 1$ GW:

$$E_{\text{src}} \approx 18.9 \text{ TJ} \approx 5240 \text{ MWh}, \quad t_{\text{src}} \approx 5.2 \text{ h}.$$

So a 1 GW CHARM bus is an **energy-throughput** match for Shuttle-class SSTO only if the climb/insert lasts hours at high average power—or peak $P_\star$ is used with κ_E toward the low end and a lighter m_{ins} .

ISS note. Matching ISS inclination adds plane-change Δv . Fold into vacuum Δv margin (§5) rather than into $\Delta \varepsilon$.

4 Flight regimes: free air → scarce air → water

Define cutoff speed v_{ab} and altitude h_{ab} where intakes seal and water thrust begins.

Regime	Density	Reaction mass	Plant → thrust path
R1 Free air	Troposphere / lower stratosphere	Free ingested air	Electricity → fan / bypass

Regime	Density	Reaction mass	Plant → thrust path
R2 Scarce air	Upper atmosphere	Free ram air	Electricity + heat → ram / plasma accelerator
R3 Vacuum	Exo-atmosphere / LEO insert	Carried water	Electricity / DEC → water plasma jet

Energy split (schematic):

$$E_{\text{src}} = E_{\text{R1}} + E_{\text{R2}} + E_{\text{R3}} + E_{\text{hotel}} + E_{\text{loss}}.$$

With free reaction mass in R1–R2, plant energy primarily raises vehicle mechanical energy and pays drag:

$$P_{\text{prop}}(t) \approx \frac{D(t) v(t)}{\eta_p(t)}, \quad \int P_{\text{prop}} dt \subset E_{\text{R1}} + E_{\text{R2}}.$$

In R3, jet power for exhaust speed v_e and mass flow $\dot{m}_w$ is

$$P_{\text{jet}} = \frac{1}{2} \dot{m}_w v_e^2, \quad T = \dot{m}_w v_e \quad \Rightarrow \quad P_{\text{jet}} = \frac{1}{2} T v_e.$$

5 Water mass as a function of whole dry mass

Water is used only in R3. Let

$$m_{\text{dry}} = m_{\text{str}} + m_{\text{pl}} + m_{\text{eng}} + m_{\text{C}} + m_{\text{bat}} + m_{\text{f}},$$

$$m_0 = m_{\text{dry}} + m_w, \quad m_{\text{ins}} = m_{\text{dry}}$$

(all water expended by insertion). Rocket equation for vacuum Δv_{vac} :

$$\Delta v_{\text{vac}} = v_e \ln \left(\frac{m_{\text{dry}} + m_w}{m_{\text{dry}}} \right) = v_e \ln \mu, \quad \mu = e^{\Delta v_{\text{vac}}/v_e}.$$

Water constraint (closed form):

$$m_w = m_{\text{dry}} \left(e^{\Delta v_{\text{vac}}/v_e} - 1 \right) = m_{\text{dry}} (\mu - 1)$$

$$m_0 = m_{\text{dry}} \mu = m_{\text{dry}} e^{\Delta v_{\text{vac}}/v_e}$$

Vacuum Δv budget:

$$\Delta v_{\text{vac}} = v_{\text{orb}} - v_{\text{ab}} + \Delta v_{g,\text{vac}} + \Delta v_{\text{steer}} + \Delta v_{\text{ISS plane}}.$$

Reference planning values:

$$v_{\text{ab}} = 3.5 \text{ km/s}, \quad \Delta v_{g,\text{vac}} + \Delta v_{\text{steer}} = 0.8 \text{ km/s}, \quad \Delta v_{\text{ISS plane}} = 0.2 \text{ km/s} \Rightarrow \Delta v_{\text{vac}} \approx 5.2 \text{ km/s}.$$

Optimistic air-breathing ($v_{\text{ab}} = 5 \text{ km/s}$, small margins) $\rightarrow \Delta v_{\text{vac}} \approx 3.5\text{--}4 \text{ km/s}$.

Exhaust speed from specific impulse:

$$v_e = I_{\text{sp}} g_0.$$

Δv_{vac}	I_{sp}	v_e	μ	m_w/m_{dry}
4 km/s	2000 s	19.6 km/s	1.226	0.226
4 km/s	3000 s	29.4 km/s	1.146	0.146
5.2 km/s	2000 s	19.6 km/s	1.302	0.302
5.2 km/s	5000 s	49.0 km/s	1.112	0.112

6 Vehicle sizing equations

6.1 Structure, crew, gear, controls, and bay

Cargo bay geometry (goal G3):

$$L_{\text{bay}} = 18.3 \text{ m}, \quad D_{\text{bay}} = 4.6 \text{ m}, \quad V_{\text{bay}} \approx \frac{\pi}{4} D_{\text{bay}}^2 L_{\text{bay}} \approx 304 \text{ m}^3.$$

Payload density check:

$$\bar{\rho}_{\text{pl}} = \frac{m_{\text{pl}}}{V_{\text{bay}}} \approx 80 \text{ kg/m}^3$$

(compatible with mixed cargo; bay remains payload volume).

Crew (Shuttle functions, single deck). Forward **flight deck**: commander and pilot **facing forward** into windows and a full control-panel wall [14,21]. Living volume: **six seats** with moderate nap recline; **waste collection system**; **galley/food**; **crew luggage / personal stowage**; **ECLSS** with explicit **oxygen and nitrogen / pressure-control tankage** (not a token rack). **Forward/port crew door (side hatch)** for terrestrial ingress only. **Airlock** oversized vs a suit-closet: dual-hatch volume on the aft cabin bulkhead facing the **cargo bay**, sized for suited egress (Shuttle middeck airlock pattern, not undersized) [21].

Landing gear and control surfaces are explicit mass lines (not buried only in a lump “structure” number):

$$\begin{aligned} m_{\text{af}} &= 7.20 \times 10^4 \text{ kg} && \text{(longer fuselage, wings, TPS, doors),} \\ m_{\text{gear}} &= 4.00 \times 10^3 \text{ kg} && \text{(nose + dual main trucks; mass bill only—see profile),} \\ m_{\text{ctrl}} &= 3.00 \times 10^3 \text{ kg} && \text{(elevons, rudder, speed-brake equiv. + actuators),} \\ m_{\text{crew}} &= 8.50 \times 10^3 \text{ kg} && \text{(ECLSS, O}_2\text{/N}_2\text{, WCS, galley, food, luggage, airlock),} \\ m_{\text{str}} &= m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}} = 8.75 \times 10^4 \text{ kg.} \end{aligned}$$

CHARM and propulsion water **do not** consume V_{bay} . Plant volume sits in a **fuselage island** (spine / aft of bay / wing carry-through), subject to

$$V_C \leq V_{\text{island}}^{\text{max}} \approx 80\text{--}150 \text{ m}^3 \quad (\text{guesstimate for pancaked Shuttle envelope}).$$

6.2 CHARM mass hole

Define island specific power on the **electrical bus**:

$$\alpha_C = \frac{P_\star}{m_C} \quad \Rightarrow \quad \boxed{m_C = \frac{P_\star}{\alpha_C}}.$$

α_C	m_C at $P_\star = 1$ GW	Comment
5 kW/kg	200 t	Heavy; fights SSTO
10 kW/kg	100 t	Stretch
15 kW/kg	67 t	Reference hole
25 kW/kg	40 t	Aggressive

Volume consistency:

$$\bar{p}_C = \frac{P_\star}{V_C} \quad \Rightarrow \quad V_C = \frac{P_\star}{\bar{p}_C}.$$

For $V_C \leq 120 \text{ m}^3$: $\bar{p}_C \geq 8.3 \text{ MW/m}^3$ bus-averaged over the island—severe packaging.

6.3 Engine and battery holes

$$m_{\text{eng}} = 1.5 \times 10^4 \text{ kg} \quad (\text{single combined-cycle string guesstimate}),$$

$$m_{\text{bat}} = 2.0 \times 10^3 \text{ kg} \quad (\text{restart} + \text{hotel}; \text{ground cart does first light}),$$

$$m_{\text{f}} = 5.0 \times 10^2 \text{ kg} \quad (p\text{-}^{11}\text{B inventory; energy-rich}).$$

6.4 Closed dry / wet mass

$$\begin{aligned} m_{\text{str}} &= m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}}, \\ m_{\text{dry}} &= m_{\text{str}} + m_{\text{pl}} + m_{\text{eng}} + \frac{P_\star}{\alpha_C} + m_{\text{bat}} + m_{\text{f}}, \\ m_{\text{w}} &= m_{\text{dry}} \left(e^{\Delta v_{\text{vac}}/v_e} - 1 \right), \\ m_0 &= m_{\text{dry}} e^{\Delta v_{\text{vac}}/v_e}, \\ m_{\text{ins}} &= m_{\text{dry}}. \end{aligned}$$

7 Solved reference vehicle (all-up mass)

Freeze:

$$\begin{aligned} P_\star &= 1 \text{ GW}, \quad \alpha_C = 15 \text{ kW/kg}, & \Delta v_{\text{vac}} &= 4.0 \text{ km/s}, \\ I_{\text{sp}} &= 2000 \text{ s}, \quad v_e = I_{\text{sp}} g_0 = 19.61 \text{ km/s}. \end{aligned}$$

Solve:

$$m_C = \frac{10^9}{1.5 \times 10^4} = 6.67 \times 10^4 \text{ kg}.$$

$$m_{\text{str}} = 72\,000 + 4\,000 + 3\,000 + 8\,500 = 8.75 \times 10^4 \text{ kg},$$

$$m_{\text{dry}} = 87\,500 + 24\,400 + 15\,000 + 66\,700 + 2\,000 + 500 = 1.961 \times 10^5 \text{ kg} \quad (196.1 \text{ t}),$$

$$\mu = e^{4000/19613} = 1.226, \quad m_w = 0.226 m_{\text{dry}} = 4.43 \times 10^4 \text{ kg} \quad (44.3 \text{ t}),$$

$$m_0 = 2.404 \times 10^5 \text{ kg} \quad (\mathbf{240 \text{ t GLOW}}),$$

$$m_{\text{ins}} = 196.1 \text{ t}.$$

Mass bill (reference):

Item	Mass
Airframe + TPS (longer OML)	72.0 t
Landing gear	4.0 t
Control surfaces + actuators	3.0 t
Crew systems (ECLSS, O ₂ /N ₂ , WCS, galley, food, luggage, airlock)	8.5 t
Payload (cargo bay)	24.4 t
Combined-cycle engine	15.0 t
CHARM island	66.7 t
Flight battery	2.0 t
$p\text{-}^{11}\text{B}$ / proton fuel (+ tankage)	0.5 t
Water (vacuum propellant)	44.3 t
GLOW	240 t

Energy at this mass:

$$E_{\text{orb}} = 6.49 \text{ TJ} = 1800 \text{ MWh}, \quad E_{\text{src}}(\kappa_E = 3) = 5410 \text{ MWh}, \quad t_{\text{src}}(1 \text{ GW}) = 5.4 \text{ h}.$$

Sensitivity (same $\Delta v_{\text{vac}}, I_{\text{sp}}; m_{\text{str}}$ fixed):

α_C	m_C	m_{dry}	m_w	m_0
10 kW/kg	100 t	229 t	52 t	281 t
15 kW/kg	67 t	196 t	44 t	240 t
25 kW/kg	40 t	169 t	38 t	208 t

Municipal takeoff weight $\sim 210\text{--}280 \text{ t}$ is heavy vs airliners but in the large-military / 747-class band—not a Citation.

8 Constraints on the CHARM power plant

Summarize as a requirement vector $\mathcal{R}_C$:

8.1 Power and mass

$$P_{\text{bus}}(t) \geq P_{\text{prop}}(t) + P_{\text{hotel}}(t), \quad \max P_{\text{bus}} = P_{\star} = 1 \text{ GW},$$

$$m_{\text{C}} = \frac{P_{\star}}{\alpha_{\text{C}}}, \quad \alpha_{\text{C}} \geq 15 \text{ kW/kg (reference; } \geq 10 \text{ kW/kg hard floor for SSTO)}.$$

$$V_{\text{C}} \leq 120 \text{ m}^3, \quad \bar{p}_{\text{C}} = P_{\star}/V_{\text{C}} \geq 8 \text{ MW/m}^3.$$

8.2 Fuel and ash

$$\dot{N}_{p^{11}\text{B}} = \frac{P_{\text{fusion}}}{8.7 \text{ MeV}}, \quad m_{\text{f}}(t_{\text{mission}}) \ll m_{\text{w}}.$$

Ash (He) strained per CHARM multi-chamber design; DEC captures charged-product free energy.

8.3 DEC

$$P_{\text{bus}} = \eta_{\text{DEC}} P_{\alpha, \text{ordered}} + \eta_{\text{th}} P_{\text{thermal reject}}, \quad \eta_{\text{DEC}} \gtrsim 0.4\text{--}0.7 \text{ (design target band)}.$$

Thermal reject (X-ray, wall, inefficiencies) must be dumped to:

- air path (R1–R2 heat exchanger / microwave), and/or
- island bath $\rightarrow$ compact turbine, and/or
- radiators only after exo-atmospheric (limited area).

8.4 Restartable in space

$$E_{\text{restart}} \leq \eta_{\text{bat}} m_{\text{bat}} e_{\text{bat}},$$

with $m_{\text{bat}} = 2 \text{ t}$ reference and $e_{\text{bat}} \sim 0.15\text{--}0.25 \text{ kWh/kg} \Rightarrow 300\text{--}500 \text{ kWh}$ class store—enough for **pilot-chamber relight** + **cascade**, not for ascent.

$$\text{Ground cart: } P_{\text{cart}} \geq P_{\text{light-off}}, \quad \text{not carried.}$$

Doctrine: continuous burn nominally; pilot-string kindling from m_{bat} if segmented; glide if relight fails.

8.5 Continuous operation / no beam farm

Recirculating power is RF walls, rotation, magnets, vacuum, cryo:

$$P_{\text{recirc}} = f_{\text{r}} P_{\text{fusion}}, \quad P_{\text{bus}} = (1 - f_{\text{r}}) P_{\text{fusion, net to bus}}.$$

Design intent: f_{r} small enough that 1 GW **bus** does not require multi-MW neutral beams.

8.6 Municipal and flight safety

- No tritium breeding inventory.
- Shield so that ramp and cabin doses meet civil constraints with plant running in fan mode.
- Single-string plant: accept engine-out $\equiv$ plant-out $\rightarrow$ glide.

8.7 How CHARM is lit (and how much energy)

CHARM is **not** lit with a neutral-beam farm. Published architecture lights a **rotating open-field mirror** with **species-separated chambers** and **RF / ponderomotive walls** [1,8,9]. A practical light-off sequence for this vehicle is:

Step	Action	Power plant elements
0	Evacuate, interlocks, ground-cart connect	Vacuum, controls
1	Energize mirror / chamber magnets	Magnet PSU $\leftarrow$ cart or battery
2	Spin up plasma rotation (centrifugal boron trap)	Rotation drive; energy later recoverable in principle [1]
3	Raise RF one-way / ponderomotive barriers	RF units
4	Inject protons and boron into separated regions	Fuel injectors + p / ^{11}B tanks
5	Establish fusion cell; route alphas to heat-exchange / DEC	DEC electrodes / wave couplers
6	Disconnect cart; bus takes hotel + propulsion	Continuous burn doctrine

Energy scale (engineering estimate — CHARM papers do not publish a flight kWh BOM):

Magnet + RF + rotation spin-up for a segmented 1 GW island is treated as 50–200 kWh class to first useful plasma (seconds–minutes of MW-class RF/magnet draw), not **MWh-class beams**. That is why

$$E_{\text{restart}} \leq 300\text{--}500 \text{ kWh} \quad (m_{\text{bat}} = 2 \text{ t})$$

is booked for **on-orbit relight**, while **first light on Earth** uses the ground cart. One-way RF walls can be energetically expensive if overused [1]; the vehicle CONOPS keeps barriers as needed for separation, not as a continuous full-power sink that eats the 1 GW bus.

Pilot-string kindling: battery (or cart) lights a fraction of chambers $\rightarrow$ DEC/RF bus cascades the rest.

9 Combined-cycle engine (detail)

One propulsion string with mode variable $\sigma \in \{1, 2, 3, 4\}$.

9.1 Mode map

σ	Name	Mach / ambient	Mass flow	Thrust model (0D)
1	Electric fan	$0 \leq M \lesssim 0.9$	$\dot{m}_{\text{air}}$	$T = \eta_f P_{\text{bus}} / v_{\text{jet}}$
2	Thermal ram	$0.9 \lesssim M \lesssim 5$	ram $\dot{m}_{\text{air}}(M, \rho)$	heat addition $\Delta h = \eta_h P_{\text{th}} / \dot{m}$
3	Plasma / MHD	$M \gtrsim 5$, scarce ρ	small $\dot{m}_{\text{air}}$	$T = \eta_{\text{MHD}} P_{\text{bus}} / v_{\text{eff}}$
4	Water plasma	intakes sealed	$\dot{m}_{\text{w}}$	$T = \dot{m}_{\text{w}} v_e$, $P_{\text{jet}} = \frac{1}{2} \dot{m}_{\text{w}} v_e^2$

Switching laws (schematic):

$$\sigma = \begin{cases} 1 & \rho > \rho_1, \ M < M_{12}, \\ 2 & \rho > \rho_2, \ M_{12} \leq M < M_{23}, \\ 3 & \rho_3 < \rho \leq \rho_2, \ M \geq M_{23}, \\ 4 & \rho \leq \rho_3 \text{ or intakes commanded sealed.} \end{cases}$$

9.2 Fan mode (municipal)

$$P_{\text{fan}} = \eta_m^{-1} T v_{\text{to}}, \quad T \geq (T/W)_{\min} m_0 g_0.$$

For $(T/W)_{\min} = 0.25$, $m_0 = 240 \text{ t}$, $v_{\text{to}} = 80 \text{ m/s}$, $\eta_m = 0.85$:

$$T \approx 0.57 \text{ MN}, \quad P_{\text{fan}} \approx 54 \text{ MW}$$

—well below $P_\star$; municipal segment is **not** the power driver.

9.3 Ram / plasma modes

Ram drag and net thrust:

$$D_{\text{ram}} = \frac{1}{2} \rho v^2 C_D S, \quad P_{\text{need}} \approx \frac{(D_{\text{ram}} - T_{\text{excess}}) v}{\eta_p}.$$

Peak P_{need} near mid-hypersonic climb sets the 1 GW requirement (design goal G9).

9.4 Water mode

$$\dot{m}_{\text{w}}(t) = \frac{T(t)}{v_e}, \quad \int_0^{t_{\text{vac}}} \dot{m}_{\text{w}} dt = m_{\text{w}},$$

$$P_{\text{bus},4}(t) \geq \frac{1}{2\eta_{\text{jet}}} \dot{m}_{\text{w}} v_e^2.$$

At $T = 100 \text{ kN}$, $v_e = 20 \text{ km/s}$: $P_{\text{jet}} = 1 \text{ GW}$ — vacuum thrust is power-limited; insertion is a **long burn**, not a Shuttle SSME sprint.

9.5 Physical envelope

- One rectangular / axisymmetric nacelle blended to wing root or aft fuselage.
- Variable inlet; **airlock shutters** for R3 and reentry.
- Shared nozzle for modes 2–4; fan bypass duct for mode 1.
- Magnetic bearings / dry lubrication where vacuum survival matters.
- $m_{\text{eng}} = 15\text{ t}$ reference includes motors, microwave/MHD hardware, pumps, and structure.

10 Vehicle layout schematics

Crew volume flattens the **Space Shuttle crew module** from two decks to **one** [14,21], then **stretches** the pressurized nose so life support and a suited airlock are not cartoon-thin. Reference overall length $L \approx 52\text{ m}$ (was 45 m): the extra length pays for O_2/N_2 tankage, luggage, a larger airlock, and dedicated **flight-battery** and $p\text{-}^{11}\text{B}$ /**proton fuel-tank** bays forward of CHARM. ECLSS = **Environmental Control and Life Support System**. The **interior floorplan omits landing gear** (gear remains on the exterior profile and in the mass bill). Depth $\approx 6.5\text{--}7\text{ m}$, span $\approx 28\text{ m}$.

10.1 Top-down floorplan

Station (m)	Bay	Contents
0–11	Crew module	Forward-facing CDR/PLT + flight deck; four more nap-recline seats (six total); WCS; galley; luggage stowage ; ECLSS + O_2/N_2 / pressure tanks
—	Ground door	Forward/port crew door (side hatch) — terrestrial ingress only
11–15	Airlock	Suited-crew airlock ($\sim 2.5\text{ m}$ class clear), aft bulkhead facing into cargo bay
15–33.3	Cargo	$18.3\text{ m} \times 4.6\text{ m}$ payload bay (no reactors)
33.3–35.5	Battery	Flight battery $\approx 2\text{ t}$ (restart / hotel)
35.5–37.5	Fusion fuel	Proton / ^{11}B feed tanks + plumbing (low mass, real volume)
37.5–45	CHARM	Reactor island ($\lesssim 120\text{ m}^3$, 67 t)
45–49	Water	$\approx 44\text{ t H}_2\text{O}$

Station (m)	Bay	Contents
49–52	Engine	Combined-cycle nacelle + nozzle
Wings	Controls	Elevons, rudder (gear not drawn on this figure)

Doors (Shuttle pattern). (1) **Side/forward crew door** — runway/ground only; (2) **airlock** — on-orbit cabin $\leftrightarrow$ cargo bay / vacuum for suited operations [21].

10.2 CHARM power-plant schematic

10.3 Profile view (to scale)

Exterior OML target $L \approx 52$ m to match the stretched floorplan. Landing gear is shown on the **profile** figure (not on the interior floorplan).

11 Systems checklist (equations $\rightarrow$ constraints)

Step	Governing relations	Binding output
Goals	G1–G9 + single-deck crew	Shuttle bay + municipal SSTO to ISS
LEO energy	$E_{\text{orb}} = m_{\text{ins}}\Delta\varepsilon$, $E_{\text{src}} = \kappa_E E_{\text{orb}}$	~ 1.8 GWh orbital @ 196 t; ~ 5.4 GWh source @ $\kappa_E = 3$
Regimes	R1–R3 mass-flow logic	Free air $\rightarrow$ scarce air $\rightarrow$ water
Water	$m_{\text{w}} = m_{\text{dry}}(e^{\Delta v_{\text{vac}}/v_e} - 1)$	$\sim 23\%$ of dry mass @ ref
Structure	$m_{\text{str}} = m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}}$	87.5 t incl. gear/controls/ECLSS+O ₂ /luggage/airlock
CHARM	$m_{\text{C}} = P_{\star}/\alpha_{\text{C}}$, DEC, restart	67 t @ 15 kW/kg
Light-off	Magnets + RF + rotation; no NBI	50–200 kWh est.; cart / 2 t battery
Solve	$m_0 = m_{\text{dry}}e^{\Delta v_{\text{vac}}/v_e}$	$m_0 \approx 240$ t; $L \approx 52$ m
Engine	Mode map σ ; $P_{\text{fan}} \ll P_{\star}$; $P_{\text{jet}} = \frac{1}{2}Tv_e$	1 GW sizes R2–R4

12 Imputed CHARM plant specs, gap to present design, and unobtainiums

12.1 Specs this vehicle imputes to CHARM

The SSTO solve does not invent new plasma physics; it **back-solves** a power island that must fit the airframe. Reference imputed plant:

Quantity	Imputed requirement	Source in this note
Fuel	Continuous $p\text{-}^{11}\text{B}$	G8; §2

Quantity	Imputed requirement	Source in this note
Architecture	Multi-chamber rotating mirror; species separation; ash strain; DEC	§2, §8, §10.2; [1,8–11]
Electrical bus peak	$P_{\star} = 1 \text{ GW}$	G9; Phase B/C power
Mission source energy	$\sim 5 \text{ GWh}$ class per ascent ($\kappa_E \sim 3$)	§3, §7
Island mass	$m_C \approx 67 \text{ t}$	$\alpha_C = 15 \text{ kW/kg}$
Specific power (bus)	$\alpha_C \geq 15 \text{ kW/kg}$ (floor $\sim 10 \text{ kW/kg}$)	§6.2, §8.1
Island volume	$V_C \lesssim 120 \text{ m}^3$	Fuselage bay aft of cargo
Volumetric power	$\bar{p}_C \gtrsim 8 \text{ MW/m}^3$	$P_{\star}/V_C$
DEC	$\eta_{\text{DEC}} \sim 0.4\text{--}0.7$ on ordered α / wave channel	§8.3
Aux heating	RF + rotation + magnets; no multi-MW NBI farm	§8.5, §8.7
Light-off energy	$\sim 50\text{--}200 \text{ kWh}$ to useful plasma (est.)	§8.7
Space restart	Pilot-string from $\sim 2 \text{ t}$ battery ($\sim 300\text{--}500 \text{ kWh}$)	§5, §8.4
Duty	Continuous burn through climb/insert; throttleable bus	§5
Environment	Flight loads, TPS-adjacent thermal, municipal dose with fan-mode plant	G6, §8.6

12.2 Where published CHARM stands today

Relative to that invoice, the Fisch / ARPA-E / Pale Blue line today is a **physics and IP program**, not a flight power plant [1,2]:

Gate	Present CHARM (public)	Vehicle need
Fuel / kinetics	Strong papers on hybrid fast–thermal $p\text{-}^{11}\text{B}$, alpha channeling, ash poisoning [3–7]	Same fuel family — aligned in intent
Chambering	Architecture + patent filings for separated reactants, ponderomotive walls, open-trap HV, differential confinement [1,8–11]	Must work together in one island
Lawson / Q	Component studies and 0D balances; integrated self-consistent power-positive reactor not demonstrated [1]	Net bus power after recirculating RF/rotation
Hardware	No public GW-class (or even pilot) CHARM machine; software-first / early company path	Flyable 67 t island
DEC	Themes in theory and open-trap HV patents [1,10]	Flight-qualified η_{DEC} at GW
Mass / volume	Not published as kW/kg or MW/m ³ plant envelopes	15 kW/kg , $\gtrsim 8 \text{ MW/m}^3$
Operations	Lab/site thinking	Continuous ascent, g-load, restart, airport licensing

Gap in one line: CHARM has a credible **aneutronic architecture story**; this vehicle

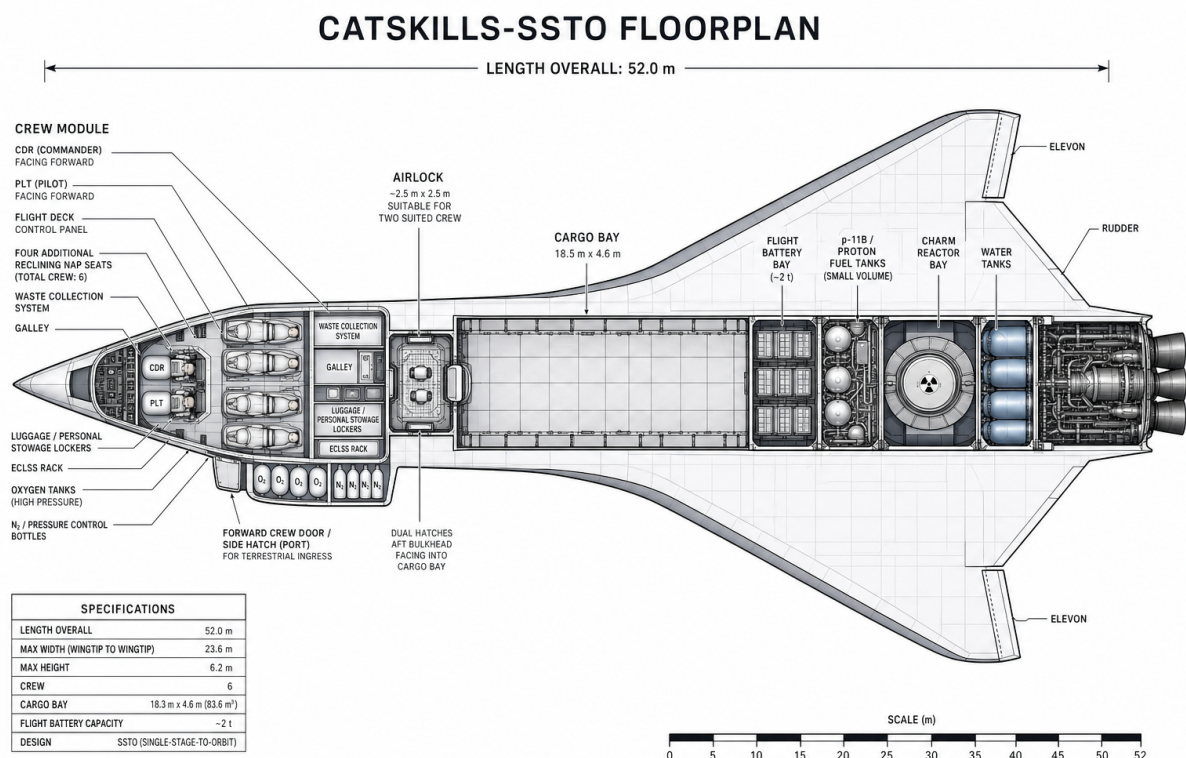


Figure 1 AI image prompt (source artifact). NASA-style top-down engineering floorplan ONLY (no landing gear drawn), elongated SSTO ~52m, blueprint gray/white. Delta wing outline with elevons/rudder labeled lightly. Forward CREW MODULE: CDR and PLT facing forward at FLIGHT DECK control panel; four more reclining nap seats (six total); WASTE COLLECTION SYSTEM (toilet); GALLEY; LUGGAGE / PERSONAL STOWAGE lockers; ECLSS rack; OXYGEN TANKS and N₂/pressure control bottles clearly labeled; FORWARD CREW DOOR / SIDE HATCH on port for terrestrial ingress. Large AIRLOCK ~2.5m square suitable for two suited crew, aft bulkhead facing into CARGO BAY with dual hatches. CARGO BAY 18.3×4.6 m. Then FLIGHT BATTERY bay (~2 t). Then p-11B / PROTON FUEL TANKS (small volume, labeled). Then CHARM REACTOR BAY. WATER TANKS. COMBINED-CYCLE ENGINE. Title CATSKILLS-SSTO FLOORPLAN. Scale bar ~52 m. No wheels, no gear trucks. Clean callouts.

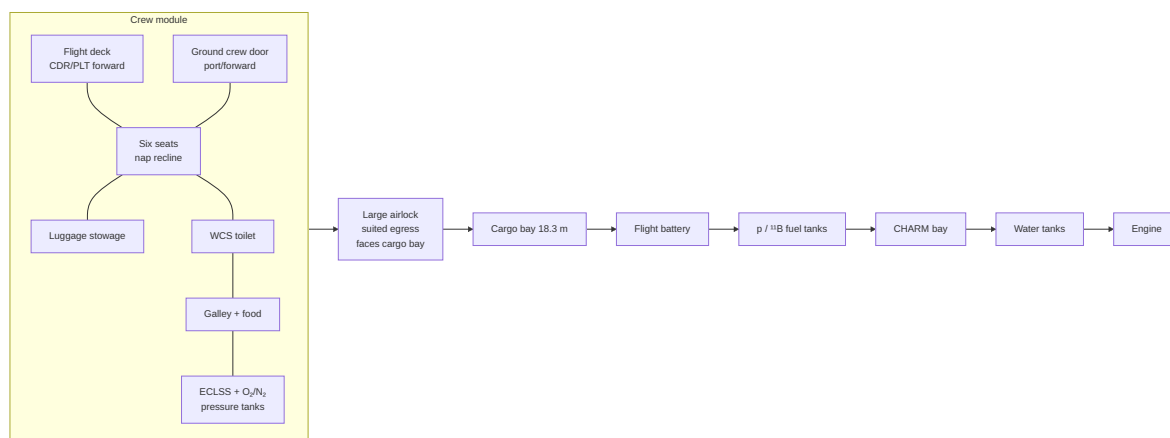


Figure 2 Top-down floorplan.

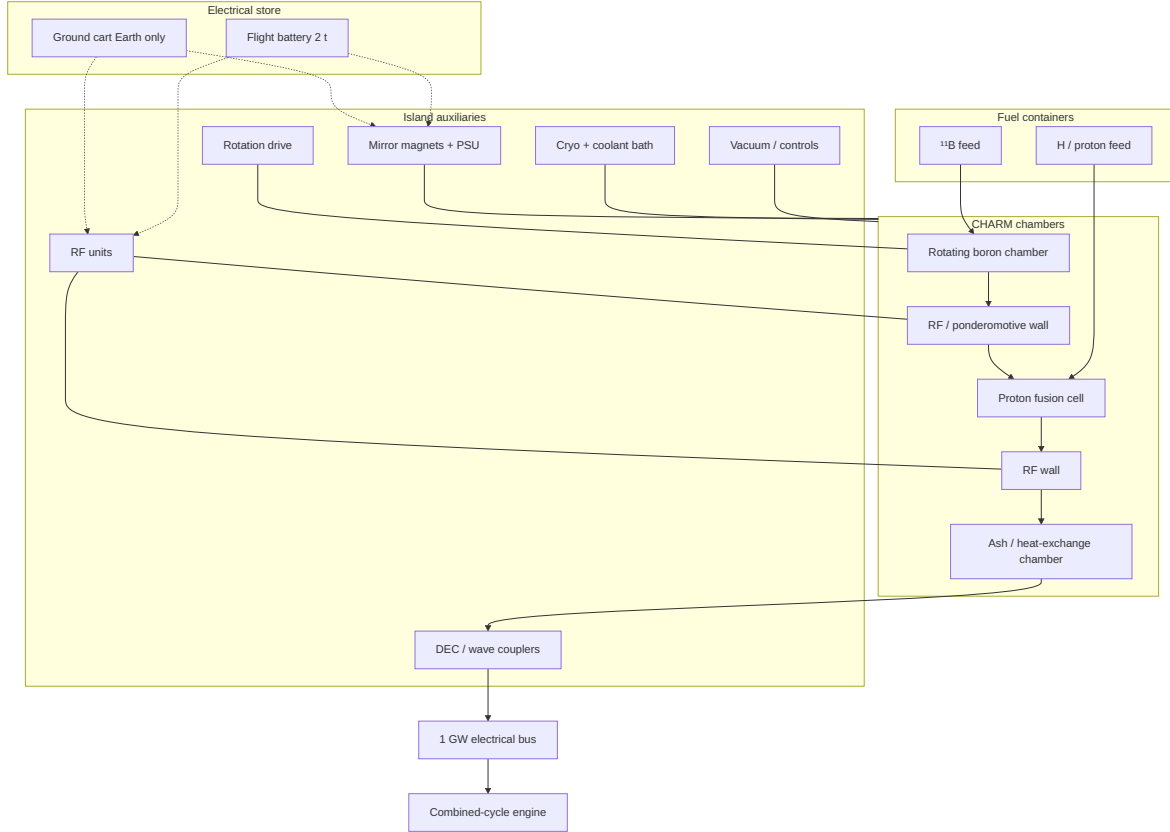


Figure 3 CHARM power-plant schematic.

needs a **closed, flight-packaged, gigawatt, high specific-power product** that does not yet exist on paper as an engineered BOM.

12.3 Unobtainiums in the gap

Call a requirement **unobtainium** if it is not implied by present CHARM results and would break the SSTO close if false. Ranked for this airframe:

1. **Integrated $p\text{-}^{11}\text{B}$ power balance with chambered species + ash removal + tolerable synchrotron/bremsstrahlung simultaneously** — components suggest feasibility; full self-consistency is explicitly still open [1].
2. **Centrifugal / rotating-mirror differential confinement at useful density and confinement time**, including rotation sustainment **without intolerable wall voltage drops** [1].
3. **Selective RF / ponderomotive “one-way” walls** that regulate ion traffic at acceptable recirculating power (slides note one-way walls can be energetically costly if overused) [1,9].
4. **Wave-mediated ash extraction / alpha channeling into protons (or DEC)** fast enough that helium does not poison the cell [3,6].

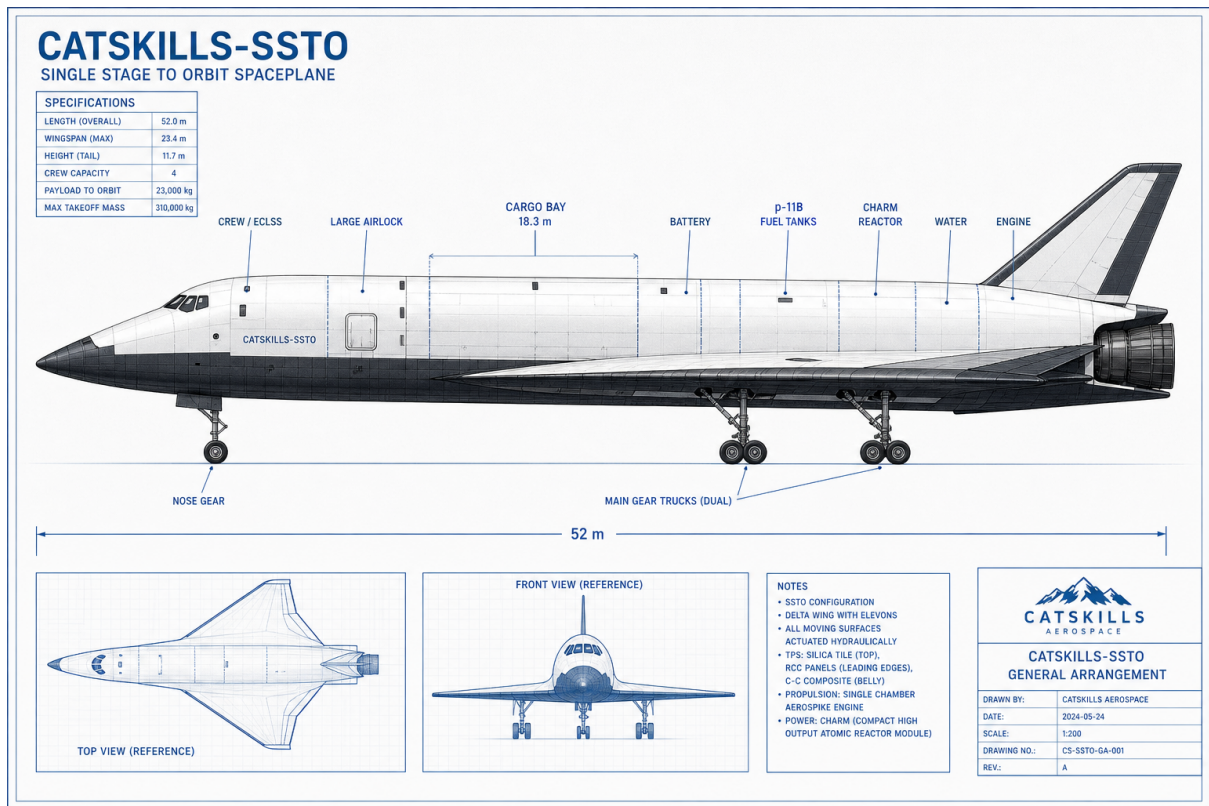


Figure 4 AI image prompt (source artifact). Technical side-profile exterior of sleek SSTO spaceplane ~52m long (longer nose than Shuttle), white upper, dark TPS belly, delta wing elevons, single vertical tail. Landing gear EXTENDED: nose gear and dual main gear trucks clearly visible and labeled. Bay labels along fuselage: CREW/ECLSS, LARGE AIRLOCK, CARGO BAY 18.3m, BATTERY, p-11B FUEL TANKS, CHARM REACTOR, WATER, ENGINE. Scale bar 52 m. Blueprint style, CATSKILLS-SSTO. Engineering drawing.

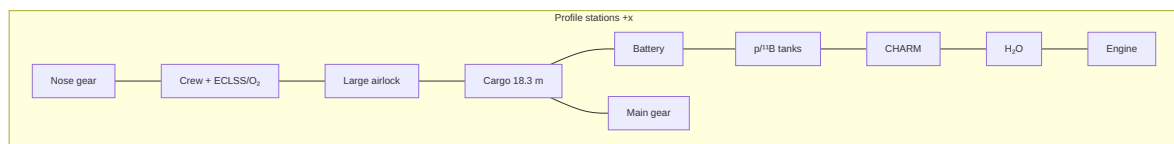


Figure 5 Profile view (to scale).

5. **Ultra-high DC / open-field electrode structures** that survive α and X-ray loads while feeding a GW bus [10].
6. **Flight specific power** $\alpha_C \sim 15 \text{ kW/kg}$ and $\bar{p}_C \gtrsim 8 \text{ MW/m}^3$ including magnets, RF, shield, cryo, and structure — beyond any published CHARM packaging study.
7. **Continuous GW-class operation** through a multi-hour ascent with vibration, thrust-vector loads, and thermal transients — not part of the present ARPA-E scope.
8. **Pilot-string light-off / space restart** at 50–200 kWh class — engineering estimate only; not a CHARM experimental result.

Items 1–5 are **physics/architecture unobtainiums** shared with any CHARM plant. Items 6–8 are **aerospace packing and CONOPS unobtainiums** imposed by SSTO. If 1–5 fail, no amount of airframe cleverness saves the mission; if 1–5 hold but 6–8 fail, CHARM may still be a ground plant while this spaceplane does not close.

12.4 How to read the rest of this paper

Sections 1–11 are a **requirements mirror** held up to CHARM: they say what a successful bottle must look like to fly Shuttle-class cargo from a municipal runway to ISS altitude. They are **not** a claim that Pale Blue / Princeton has those numbers. Closing the gap is future plasma physics, materials, and packaging work—tracked against the unobtainium list above.

13 Conclusion

Design goals fix a **Shuttle-style SSTO** with a **real cargo bay**, municipal runway ops, and ISS-altitude LEO. The pressurized nose is **lengthened** ($\approx 52 \text{ m OML}$) so the cabin can hold a real **ECLSS with O₂/N₂, luggage, a suited-size airlock** into the bay, plus visible **battery** and **fusion-fuel tank** bays. Crew: **forward-facing flight deck, six nap-recline seats, ground crew door, aft airlock into cargo. Landing gear** (4 t) and **control surfaces** (3 t) sit in the mass bill (gear drawn on the profile, not the floorplan). The MWh budget is linear in inserted mass: about 9.2 kWh **per kg** orbital, times $\kappa_E \sim 2\text{--}4$ at the CHARM bus. Water scales only with **dry mass** and vacuum $\Delta v/I_{\text{sp}}$. Closing $P_\star = 1 \text{ GW}$ at $\alpha_C = 15 \text{ kW/kg}$ yields a reference vehicle of about 196 t **dry**, 44 t **water**, 240 t **GLOW**, with CHARM as a 67 t, $\lesssim 120 \text{ m}^3$ island using **DEC**, lit by **magnets + RF + rotation** (50–200 kWh class light-off; space restart from a 2 t battery), and a **single combined-cycle engine** spanning fan $\rightarrow$ ram/plasma $\rightarrow$ water jet. Layout figures (§10) show planform, plant, and profile. Those are the numbers the in-silico CHARM plant and engine maps must satisfy.

14 Acknowledgments

CHARM denotes the chambered aneutronic rotating-mirror architecture developed in the Princeton / ARPA-E economical $p\text{-}^{11}\text{B}$ program and discussed toward Pale Blue Fusion. This vehicle sketch is an independent systems exercise and does not speak for that program.

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Working draft for arxiv-style systems sizing. Not a license to operate a reactor.